
Probability

CS5491: Artificial Intelligence
ZHICHAO LU

Content Credits: Prof. Wei's CS4486 Course
and Prof. Boddeti's AI Course



TODAY

Random Variables

Joint and Marginal Probabilities

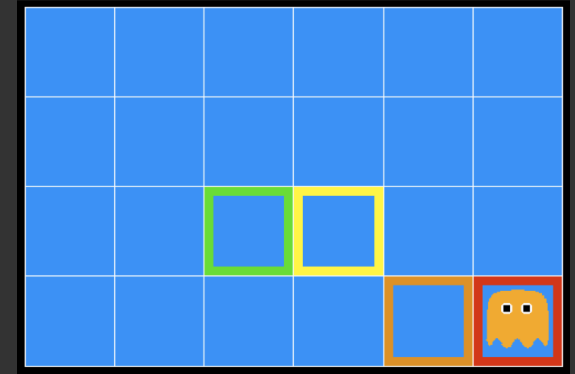
Conditional Probabilities

Product Rule, Chain Rule, Bayes Rule



INFERENCE IN GHOSTBUSTERS

A ghost is in the grid somewhere

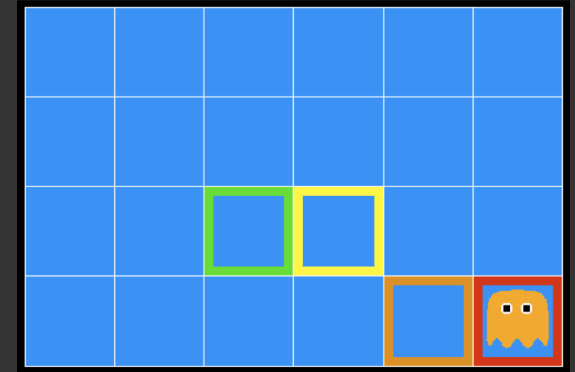


INFERENCE IN GHOSTBUSTERS

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Sensor readings tell how close a square is to the ghost

- On the ghost: red
- 1 or 2 away: orange
- 3 or 4 away: yellow
- 5+ away: green

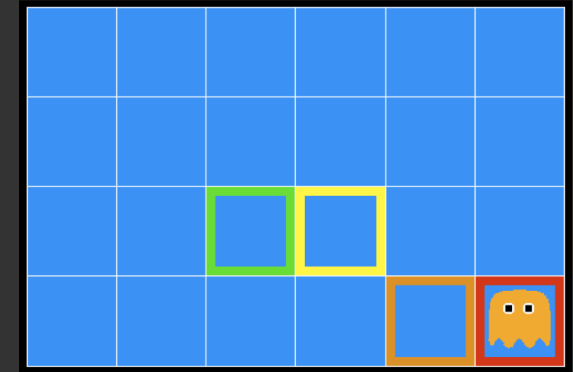


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Sensors are noisy, but we know $P(\text{color}|\text{distance})$

$pred(red 3)$	$pred(orange 3)$	$pred(yellow 3)$	$pred(green 3)$
0.05	0.15	0.50	0.30



UNCERTAINTY

General situation:

0.11	0.11	0.11
0.11	0.11	0.11
0.11	0.11	0.11

0.17	0.10	0.10
0.09	0.17	0.10
0.01	0.09	0.17

0.01	0.01	0.03
0.01	0.05	0.05
0.01	0.05	0.81



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General situation:

- **Observed variables (evidence):** Agent knows certain things about the state of the world (e.g., sensor readings or symptoms)

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Probabilistic reasoning gives us a framework for managing our beliefs and knowledge

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RANDOM VARIABLES

A random variable is some aspect of the world about which we (may) have uncertainty

- R = Is it raining?
- T = Is it hot or cold?
- D = How long will it take to drive to work?
- L = Where is the ghost?



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Like variables in a CSP, random variables have domains

- $R \in \{false, true\}$ (often written as $\{-r, +r\}$)
- $T \in \{hot, cold\}$
- $D \in [0, \infty)$
- $L \in$ possible locations, maybe $\{(0, 0), (0, 1), \dots\}$



PROBABILITY DISTRIBUTIONS

Associate a probability with each value



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Temperature $P(T)$

T	P
hot	0.5
cold	0.5



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Associate a probability with each value

Temperature $P(T)$

T	P
hot	0.5
cold	0.5

Weather $P(W)$

W	P
sun	0.6
rain	0.1
fog	0.3
meteor	0.0

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Unobserved random variables have distributions



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A distribution is a TABLE of probabilities of values



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Shorthand Notation:

$$P(\textit{hot}) = P(T = \textit{hot})$$

$$P(\textit{cold}) = P(T = \textit{cold})$$

$$P(\textit{rain}) = P(W = \textit{rain})$$

OK, *if* all domains entries are
unique



JOINT DISTRIBUTIONS

A joint distribution over a set of random variables: $X_1, X_2, \dots, X_n$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3



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Joint distribution specifies a real number for each assignment (or outcome):

$$P(X_1, \dots, X_n)$$

$$P(X_1 = x_1, \dots, X_n = x_n)$$

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Size of distribution if n variables with domain sizes d ?

- For all but the smallest distributions, impractical to write out !!

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PROBABILISTIC MODELS

A probabilistic model is a joint distribution over a set of random variables

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Probabilistic models:

- (Random) variables with domains
- Assignments are called outcomes
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- Normalized: sum to 1.0
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Constraint satisfaction problems:

- Variables with domains
- Constraints: state whether assignments are possible
- Ideally: only certain variables directly interact

<i>T</i>	<i>W</i>	<i>P</i>
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

<i>T</i>	<i>W</i>	<i>P</i>
hot	sun	T
hot	rain	F
cold	sun	F
cold	rain	T



EVENTS

An *event* is a set E of outcomes:

$$P(E) = \sum_{(x_1, \dots, x_n) \in E} P(x_1, \dots, x_n)$$

T	W	P
hot	sun	0.4
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From a joint distribution, we can calculate the probability of any event

- Probability that it is hot AND sunny?
- Probability that it is hot?
- Probability that it is hot OR sunny?

T	W	P
hot	sun	0.4
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From a joint distribution, we can calculate the probability of any event

- Probability that it is hot AND sunny?
- Probability that it is hot?
- Probability that it is hot OR sunny?

Typically, the events we care about are partial assignments, like $P(T = \text{hot})$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3



MARGINAL DISTRIBUTIONS

Marginal distributions are sub-tables which eliminate variables



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T	W	P
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hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

T	P
hot	0.5
cold	0.5



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Marginalization (summing out): Combine collapsed rows by adding

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

T	P
hot	0.5
cold	0.5

W	P
sun	0.6
rain	0.4



CONDITIONAL DISTRIBUTIONS

A simple relation between joint and conditional probabilities

- In fact, this is taken as the definition of a conditional probability



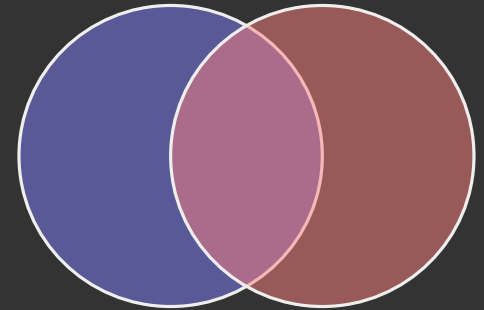
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$$P(a|b) = \frac{P(a,b)}{P(b)}$$

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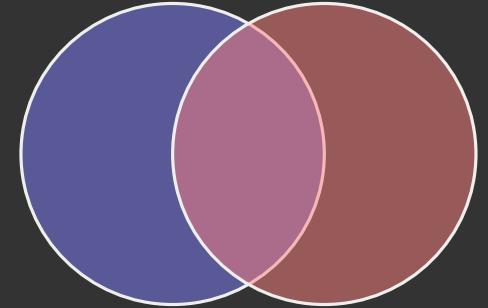


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$$P(W = s|T = c) = \frac{P(W = s, T = c)}{P(T = c)} = \frac{P(W = s, T = c)}{P(W = s, T = c) + P(W = r, T = c)} = \frac{0.2}{0.2 + 0.3} = 0.4$$



CONDITIONAL DISTRIBUTIONS

Conditional distributions are probability distributions over some variables given fixed values of others

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cold	sun	0.2
cold	rain	0.3



CONDITIONAL DISTRIBUTIONS

Conditional distributions are probability distributions over some variables given fixed values of others

$$P(W|T = \text{hot})$$

W	P
sun	0.8
rain	0.2

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

CONDITIONAL DISTRIBUTIONS

Conditional distributions are probability distributions over some variables given fixed values of others

$$P(W|T = \textit{hot})$$

W	P
sun	0.8
rain	0.2

$$P(W|T = \textit{cold})$$

W	P
sun	0.4
rain	0.6

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

NORMALIZATION TRICKS

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

W	P
sun	
rain	

NORMALIZATION TRICKS

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SELECT the joint probabilities

matching the evidence

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cold	rain	0.3

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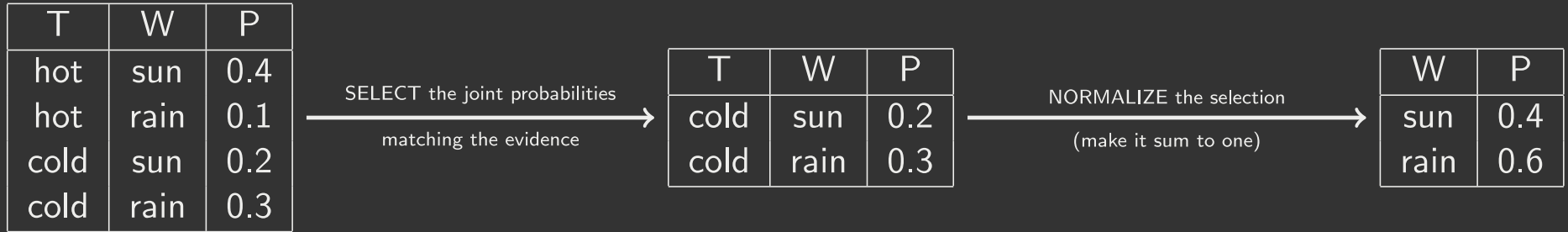
T	W	P
cold	sun	0.2
cold	rain	0.3

NORMALIZE the selection
(make it sum to one)

W	P
sun	0.4
rain	0.6



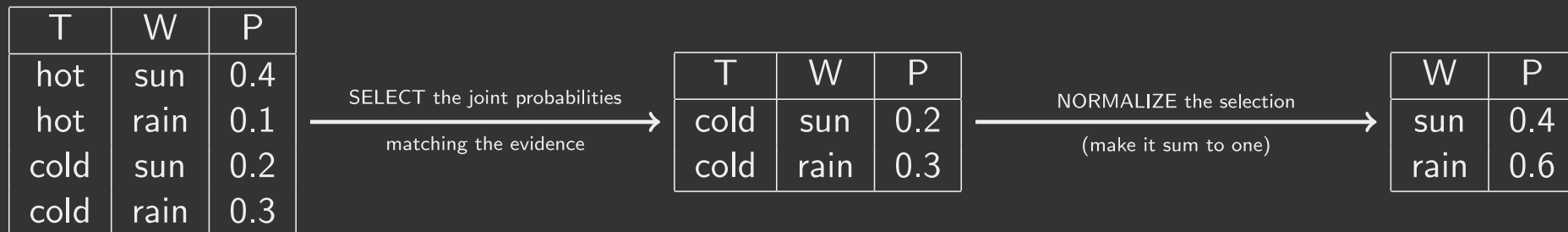
NORMALIZATION TRICKS



Why does this work?

- Sum of selection is $P(\text{evidence})$!
- $P(T = c)$ in this case

NORMALIZATION TRICKS



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- Sum of selection is $P(\text{evidence})$!
- $P(T = c)$ in this case

In general:

$$P(x_1|x_2) = \frac{P(x_1, x_2)}{P(x_2)} = \frac{P(x_1, x_2)}{\sum_{x_1} P(x_1, x_2)}$$



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(Dictionary) To bring or restore to a normal condition



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Procedure:

- Step 1: Compute $Z = \text{sum over all entries}$
- Step 2: Divide every entry by Z



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W	P		W	P
sun	0.2	$\xrightarrow[\substack{\text{NORMALIZE} \\ Z = 0.5}]{}$	sun	0.4
rain	0.3		rain	0.6

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Procedure:

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- Step 2: Divide every entry by Z

W	P
sun	0.2
rain	0.3

NORMALIZE
 $Z = 0.5$

W	P
sun	0.4
rain	0.6

X	Y	P
hot	sun	20
hot	rain	5
cold	sun	10
cold	rain	15

NORMALIZE
 $Z = 50$

X	Y	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

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Probabilistic inference: compute a desired probability from other known probabilities (e.g. conditional from joint)



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We generally compute conditional probabilities

- $P(\text{on time} | \text{no reported accidents}) = 0.90$
- These represent the agent's beliefs given the evidence



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- These represent the agent's beliefs given the evidence

Probabilities change with new evidence:

- $P(\text{on time} | \text{no accidents, 5 a.m.}) = 0.95$
- $P(\text{on time} | \text{no accidents, 5 a.m., raining}) = 0.80$
- Observing new evidence causes beliefs to be updated



INFERENCE BY ENUMERATION

Problem Setup:



INFERENCE BY ENUMERATION

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- General Case:
 - Evidence variables: $E_1, \dots, E_k = e_1, \dots, e_k$
 - Query variables: Q
 - Hidden variables: $H_1, \dots, H_r$



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Solution:

- Step 1: Select the entries consistent with the evidence
- Step 2: Sum out H to get joint of Query and evidence

$$P(Q, e_1, \dots, e_k) = \sum_{h_1, \dots, h_r} P(Q, h_1, \dots, h_r, e_1, \dots, e_k)$$



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$$P(Q, e_1, \dots, e_k) = \sum_{h_1, \dots, h_r} P(Q, h_1, \dots, h_r, e_1, \dots, e_k)$$

- Step 3: Normalize

$$Z = P(Q, e_1, \dots, e_r)$$

$$P(Q|e_1, \dots, e_r) = \frac{1}{Z} P(Q, e_1, \dots, e_r)$$



INFERENCE BY ENUMERATION

$P(W)$?

$P(W|winter)$?

$P(W|winter, hot)$?

summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20



INFERENCE BY ENUMERATION

Obvious problems:



INFERENCE BY ENUMERATION

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- Worst-case time complexity $\mathcal{O}(d^n)$



INFERENCE BY ENUMERATION

Obvious problems:

- Worst-case time complexity $\mathcal{O}(d^n)$
- Space complexity $\mathcal{O}(d^n)$ to store the joint distribution



PRODUCT RULE

Sometimes we have conditional distributions but want the joint

$$P(y)P(x|y) = P(x, y)$$
$$\Rightarrow P(x|y) = \frac{P(x, y)}{P(y)}$$



PRODUCT RULE

$$P(y)P(x|y) = P(x, y)$$

Example:

<i>W</i>	<i>P</i>
sun	0.8
rain	0.2

<i>D</i>	<i>W</i>	<i>P</i>
wet	sun	0.1
dry	sun	0.9
wet	rain	0.7
dry	rain	0.3

<i>D</i>	<i>W</i>	<i>P</i>
wet	sun	
dry	sun	
wet	rain	
dry	rain	

CHAIN RULE

More generally, can always write any joint distribution as an incremental product of conditional distributions

$$P(x_1, x_2, x_3) = P(x_1)P(x_2|x_1)P(x_3|x_2, x_1)$$

$$P(x_1, x_2, \dots, x_n) = \prod_i P(x_i|x_1, \dots, x_n)$$



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This is always true. Why?



BAYES RULE



BAYES RULE

Two ways to factor a joint distribution over two variables:

$$P(x, y) = P(x|y)P(y) = P(y|x)P(x)$$



BAYES RULE

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$$P(x, y) = P(x|y)P(y) = P(y|x)P(x)$$

Dividing, we get:

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Dividing, we get:

$$P(x|y) = \frac{P(y|x)P(x)}{P(y)}$$

Why is this at all helpful?

- Lets us build one conditional from its reverse
- Often one conditional is tricky but the other one is simple
- Foundation of many systems we'll see later.



BAYES RULE

Two ways to factor a joint distribution over two variables:

$$P(x, y) = P(x|y)P(y) = P(y|x)P(x)$$

Dividing, we get:

$$P(x|y) = \frac{P(y|x)P(x)}{P(y)}$$

Why is this at all helpful?

- Lets us build one conditional from its reverse
- Often one conditional is tricky but the other one is simple
- Foundation of many systems we'll see later.



INFERENCE WITH BAYES RULE

Example: Diagnostic probability from causal probability:

$$P(\textit{cause}|\textit{effect}) = \frac{P(\textit{effect}|\textit{cause})P(\textit{cause})}{P(\textit{effect})}$$



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- Note: you should still get stiff necks checked out !! Why?



GHOSTBUSTERS REVISITED

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- Sensor reading model: $P(R|G)$
 - Given: we know what our sensors do
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 - e.g. $P(R = \text{yellow} | G = (1, 1)) = 0.1$



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We can calculate the posterior distribution $P(G|r)$ over ghost locations given a reading using Bayes' rule:

$$P(g|r) \propto P(r|g)P(g)$$



Q & A



XKCD

